Mobile App for Detecting Artificially Ripened Mangoes Using Computer Vision and Colorimetric Analysis

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ABSTRACT

This project aims to develop an Android mobile application that detects **artificially ripened mangoes** using **computer vision** and **colorimetric analysis**. By analyzing color and texture patterns, the app helps identify **chemically ripened** fruits and provides users with a simple, affordable, and **reliable tool** to support **food safety** and **consumer awareness**.

INTRODUCTION

- The use of **chemical agents** like **calcium carbide** for ripening mangoes is a growing concern, as it poses serious **health risks** and is difficult for consumers to detect visually. Artificially ripened fruits often show **unnatural color patterns** and lack natural ripening characteristics, making it important to develop a method that can assist in distinguishing them.

- Modern **mobile technology** and **AI-based image analysis** enable the development of tools that can identify ripening abnormalities in mangoes across different varieties. Using **computer vision techniques** and **colorimetric features**, this project creates a user-friendly Android application that supports **food safety checks**, helps **consumers and officials**, and encourages **safer fruit consumption.**

METHODOLOGY

1. **Data Collection :** Images of different mango varieties were gathered at various ripening stages to create a usable dataset.
2. **Pre-processing :** The images were resized, cleaned, and converted into suitable color spaces to prepare them for analysis.
3. **Feature Extraction :** Key features such as color intensity, texture, and surface patterns were extracted to help identify signs of artificial ripening.
4. **Model Development :** A CNN model was trained to classify mangoes as naturally or artificially ripened based on the extracted features.
5. **Mobile App Integration :** The trained model was deployed using TensorFlow Lite, allowing the app to analyze a captured image and display the ripening result with a safety indication

ALGORITHMS

1. CNN Algorithm: Uses CNN to extract basic texture and color features and predict the ripening type.
2. Colorimetric Analysis Algorithm: Uses HSV color values to detect natural and unnatural yellow tones.
3. Image Segmentation: Separates the mango from the background for clearer analysis.
4. SVM Classification: Classifies the mango based on extracted features.
5. Model Optimization: Uses quantization for faster TensorFlow Lite performance.

RESULTS

App Workflow – User Interaction
1. User places the mango and captures a photo.

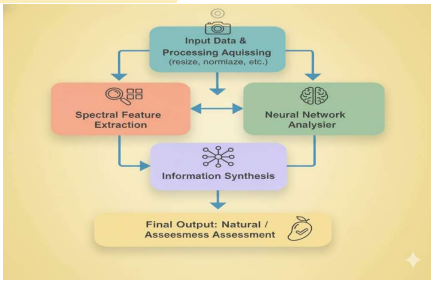
2. The app scans and analyzes the image.

3. A clear result is shown: natural or artificially ripened.

App Workflow Processing Steps:
1. The app captures and prepares the mango image.

2. Color and texture are analyzed using CNN.

3. The system combines the results and gives the ripening status.



CONCLUSION

The app accurately identifies **artificially ripened mangoes** using **CNN** and **color analysis**, helping improve **consumer safety** and support **food inspection**. The model runs efficiently on **mobile devices**, making detection quick and accessible. This project highlights the usefulness of **AI-based mobile tools** for promoting **safer food choices**.

REFERENCES

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